

Week 19 – Advanced Library & Dependency Testing

Summary:

Week 19 focused on **testing all pre-installed libraries** in the backend across multiple languages.

Key Activities:

- Verified **Python libraries** (NumPy, Matplotlib, Pandas).
- Checked **Java/C/C++ dependencies** inside Docker containers.
- Ensured **library version compatibility** and no conflicts.
- Documented any limitations or special instructions for each library.

Technologies Used: Python, Matplotlib, NumPy, FastAPI, Docker, WSL2

Outcome:

Backend fully supports all **pre-installed libraries**, ensuring smooth execution for all multi-language programs.

Images (Placeholders):

```
karmandeep@DESKTOP-DMFQ002:~/projects/Universal-Compiler$ sudo docker exec -it python-x11 bash
[sudo] password for karmandeep:
root@b8802c044823:/code# python3 test_numpy.py
Testing NumPy...
Array: [1 2 3 4 5]
Sum: 15
Shape: (5,)
NumPy is working ✓
root@b8802c044823:/code#
```